

Lab 1: KIP (Knowledge Is Power)

Benjamin Keckley, et al.

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Thomas Kennedy

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1. Introduction

This first section establishes the motivation for Knowledge Is Power (KIP) by illustrating the need for greater legal literacy among young adults. It first defines the societal problem, then identifies the characteristics of a successful educational solution, and finally introduces KIP as a means of addressing that need.

1.1. Societal Problem

Young people ages 16–19 increasingly encounter laws involving driving, digital conduct, work, school, ownership, contracts, and safety without fully understanding their inner workings and the consequences that may ensue. As they gain more independence into adulthood, this lack of legal awareness can lead to poor decisions and serious consequences. Despite this, law-related education remains largely absent from K–12 curricula, leaving young adults underprepared for real-world legal situations. Teen drivers face higher safety risks that are often linked to poor judgement and decision-making. [5.1, 5.2, 5.3, 5.4]

1.2. Needed Solution Characteristics

A learning platform should address these problems by teaching legal concepts through interactive activities that ask learners to apply laws to realistic situations rather than simply memorize definitions. To remain accessible, it should use clear and consistent language, concise explanations, mobile-friendly web design, and low-stakes practice games that students can complete independently. It should also explain the connection between each lesson and its official legal sources while introducing legal vocabulary in context so that learners can recognize practical terms used across different legal topics.

1.3. Product Introduction

KIP is a digital legal literacy platform imagined for this purpose. It is designed to teach essential, everyday laws to young adults, serving as a high-quality supplemental educational aid for parents, teachers, and school administrators alike. KIP's goal is to help young adults recognize legally-significant situations and understand the potential consequences of their choices and prepare them to make more informed decisions. The prototype will demonstrate these goals through a web-based service that supports user accounts, legal topics, interactive scenarios, progress tracking, and recommendation features based on user performance.

2. KIP Product Description

This section translates the aforementioned educational needs into the proposed design of KIP. It first describes the product's overall structure and approach to learning, then identifies the technical features and foundational capabilities that will facilitate this approach.

2.1. Key Product Features and Capabilities

KIP is structured as a web-based educational service built around interactive games, legal topics, user progress data, and source-backed legislative content. The system presents learners with games of varying designs in order to adapt to different learning styles, session lengths, and topical familiarity, then uses their performance to recommend future practice. The prototype will include two games: a short quiz centered on clearing up common legal misconceptions and a longer narrative game based on branching decisions. Its software components support account access, scenario delivery, feedback generation, content curation, source referencing, and persistent progress tracking. The goal of these features is to assess learners' current legal

understanding, strengthen their recognition of legal situations, and reinforce informed decision making through repeated low-stakes practice.

2.1.1. Solution Overview

There are three levels at which KIP supports the learner: the learning process, game practice, and adaptive progress. Learners build familiarity with legal concepts and strengthen their understanding through repeated practice. Game practice presents scenarios, evaluates responses, and provides corrective feedback, while adaptive progress records performance and uses it to prioritize future content. As illustrated in Figure 1, these levels form a feedback loop in which practice shapes future practice and contributes to broader legal understanding.

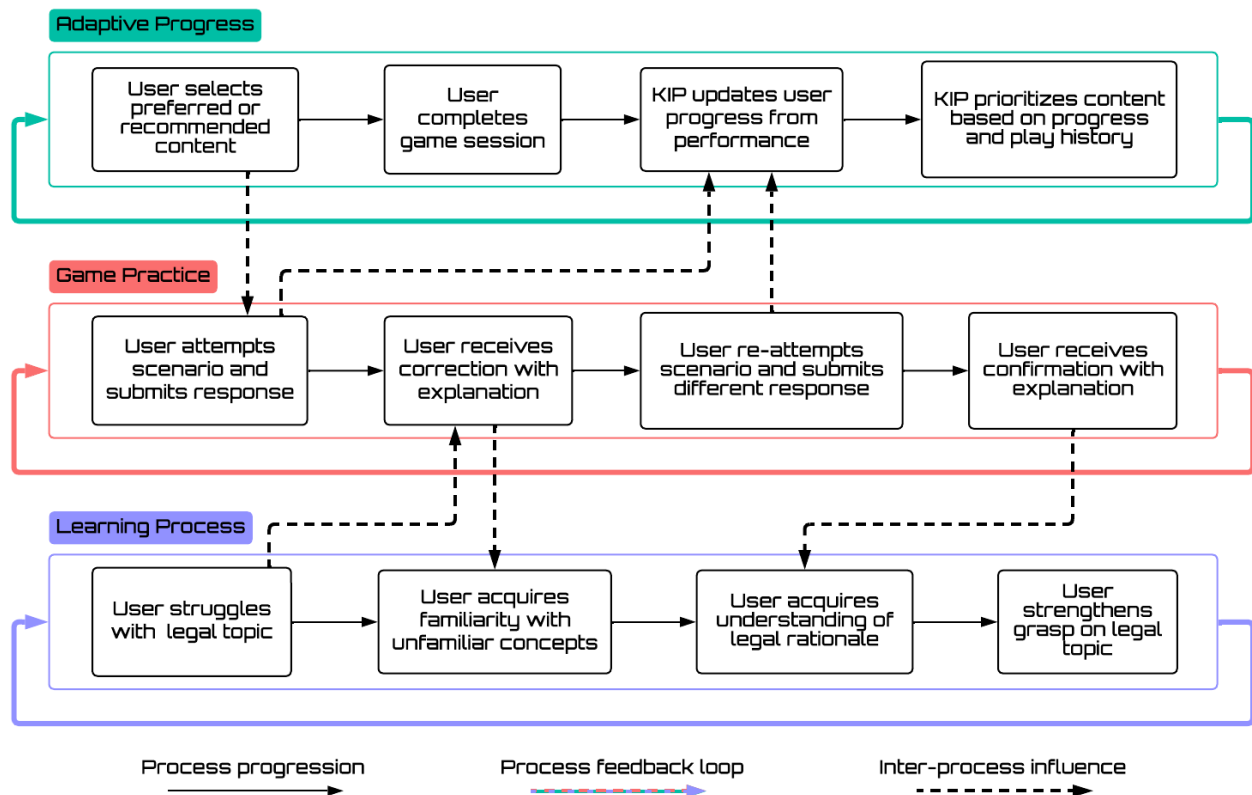


Figure 1. *Flowchart: Solution Process*

2.1.2. Goals and Objectives

KIP has three primary educational objectives: assess a learner's current understanding of practical legal topics, identify which areas require additional practice, and reinforce accurate legal reasoning through repeated engagement and feedback.

2.1.3. Interactive Learning Games

The prototype for KIP includes two games designed to exercise different forms of legal understanding.

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2.1.3.1. “Fact or Fiction?” Game

This short-form game presents learners with common legal beliefs and misconceptions, offering answers that represent “true”, “false”, or “context-dependent”, as seen in Figure 2. This approach will help learners challenge assumptions, spot legal nuance, and build foundational knowledge to prepare them for longer and more elaborate scenarios.

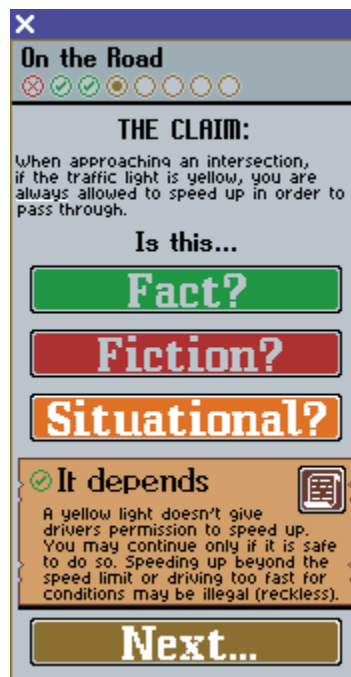


Figure 2. Mobile GUI Mockup: “Fact or Fiction?”

2.1.3.2. “Branching Narrative” Game

This multiple-choice, long-form game encourages learners to interact with people, objects, and documents to gather information before making a decision, as seen in Figure 3. Each choice provides feedback explaining its consequences, then advances the scenario to a new stage with updated interactables and a new question. This complex, reactive structure encourages learners to slow down, evaluate context, and understand which pieces of information are relevant to an outcome.



Figure 3. *Mobile GUI Mockup: “Branching Narrative”*

2.1.4. Progress Tracking and Analytics

KIP tracks a registered learner's performance and recent engagement which are used holistically to evaluate user-facing progress statistics and momentum scores. These measurements illustrate improvement, strengths, weaknesses, and participation to the learner.

2.1.4.1. User Stats

A learner can view three different standout statistics derived from their learning progress and one based on game preferences. As learners focus on one topic at a time to improve their competency in it, they will see the topic graduate from one statistic to another, showing evidence of their progress.

2.1.4.1.1. “Needs Practice” Stat

This stat identifies the topic in which a learner has shown the least competency. It provides a clear objective on which topic they could stand to gain the most progress in if they focus their efforts.

2.1.4.1.2. “Making Progress” Stat

This stat identifies the topic in which a learner has recently shown active improvement, acting as a bridge between the “Needs Practice” stat and the “Strongest Topic” stat.

2.1.4.1.3. “Strongest Topic” Stat

This stat identifies the topic in which a learner has demonstrated the highest level of competency. It provides clear evidence of their achievement, reinforcing the results of their progress and motivating continued learning.

2.1.4.1.4. “Favorite Game” Stat

This stat identifies the game a learner has played the most. While topic-oriented stats visualize learners’ progress in learning, this stat reflects their individual play preferences, giving their profiles a more personal quality that distinguishes them from each other.

2.1.4.2. User Momentum Scores

KIP tracks two major analytics to encourage learners to return frequently, represented by numerical tally scores. These scores embody a similar spirit to that of high scores in contemporary video games.

2.1.4.2.1. “Current Streak” Score

This score identifies how many days in a row a learner has logged into KIP and completed at least one game session. A high score represents a learner who engages with KIP on a consistent basis, and breaking the streak means resetting the score to zero.

2.1.4.2.2. “Weekly Practice” Score

This score identifies how many game sessions a learner has played within the last seven days. A high score represents a learner who has engaged a lot recently, and a low score indicates that they have barely engaged with KIP’s content as of late, if at all.

2.2. Major Components (Hardware/Software)

KIP's software architecture is divided into presentation, application, and data layers, as shown in Figure 4. This separation keeps user-facing interfaces distinct from application logic and persistent data storage, while defining how information moves between the major functional components.

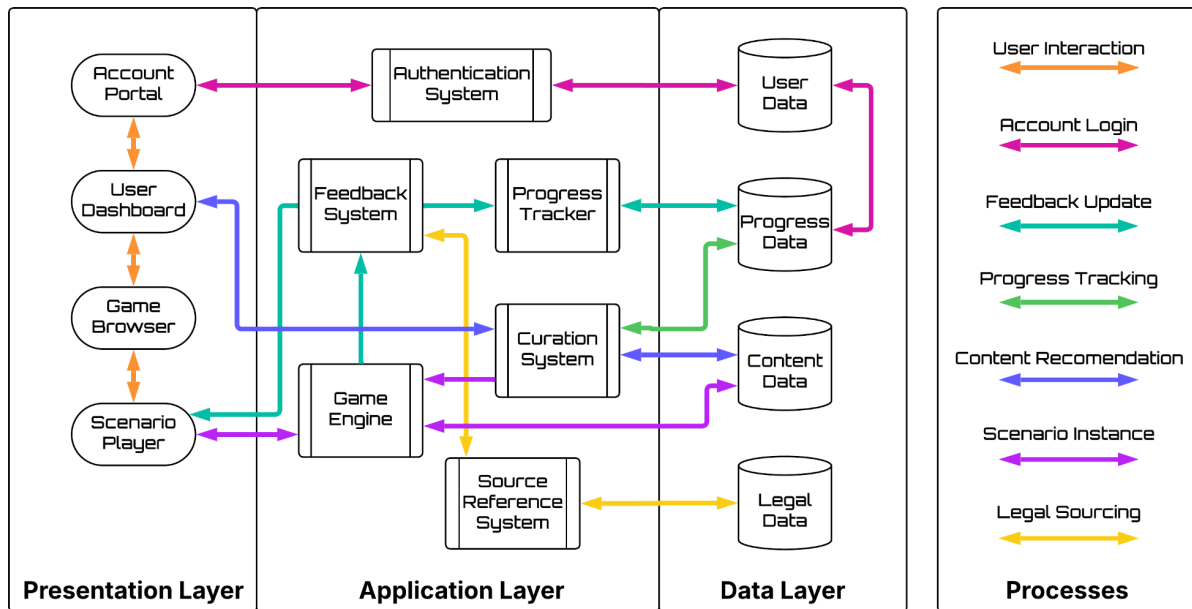


Figure 4. Diagram: *Major Functional Components*

2.2.1. Presentation Layer

The presentation layer contains KIP's user-facing interfaces and handles the display of information and graphics, as well as the collection of user input. It communicates user actions to the application layer and presents the resulting content, feedback, progress information, and recommendations returned by KIP's backend systems.

2.2.1.1. Tech Stack

KIP's frontend is built using HTML5, TypeScript, and Bootstrap. HTML5 provides the structure and organization of the application's pages, TypeScript handles client-side logic and interactivity, and Bootstrap supports the responsive layouts and reusable interface components across desktop and mobile devices.

2.2.1.2. Account Portal

This landing page allows existing learners to log in, and newcomers to create an account or play games without an account to get a taste of KIP, albeit with limited features.

2.2.1.3. User Dashboard

The page shown in Figure 5 summarizes a learner's progress in several ways. It contains a history of their game sessions, a list of their progress stats, and their current momentum scores. It also offers one recommendation for the combination of game and topic that would most strongly benefit the learner to practice next. Together, these elements give learners a centralized view of their learning activity, allowing them to recognize strengths and weaknesses, review recent performance, and identify where to focus their efforts next.



Figure 5. Desktop GUI Mockup: User Dashboard

2.2.1.4. Game Browser

The browser shown in Figure 6 displays the games available to a learner and orders them based on recent play history and overall experience with each of them. Every game entry provides information about how recently the learner played it and how frequently they have engaged with it. The browser also allows the learner to filter games by one or more selected legal topics. If a game does not offer content associated with any of the selected topics, it is excluded from the browsing results. This allows learners to narrow their search according to two important priorities: their preferred style of play and the subject matter they want to practice.

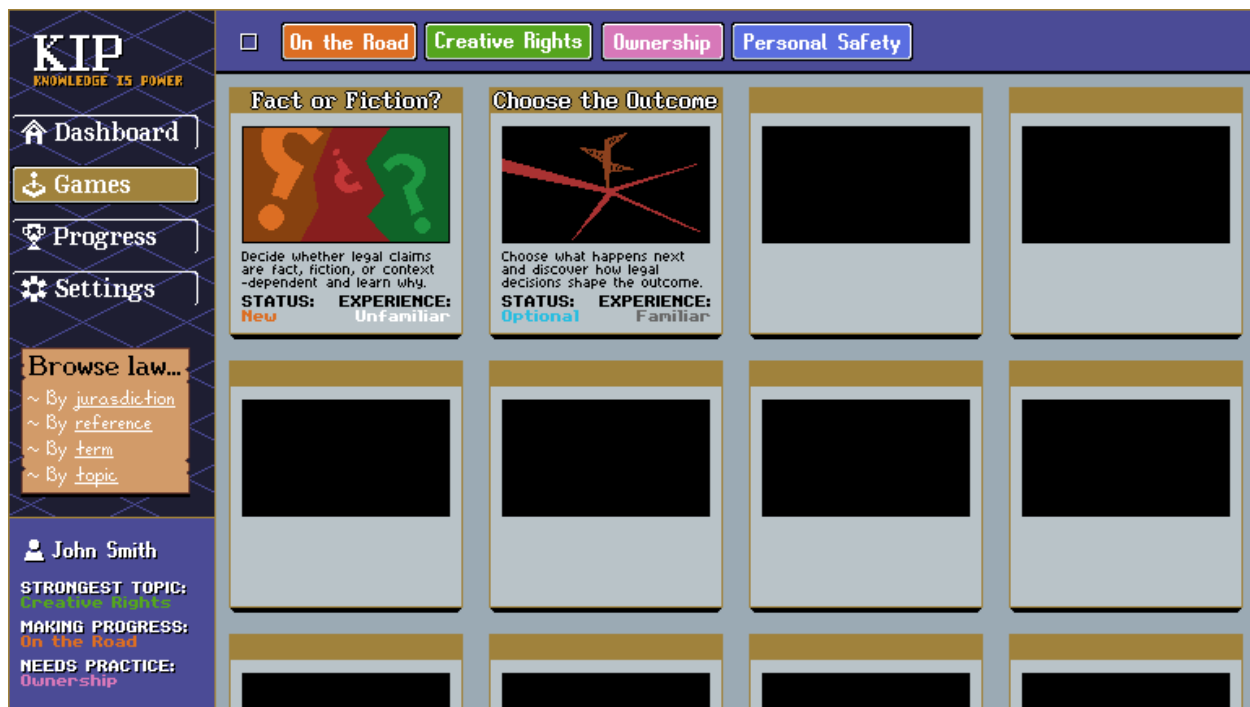


Figure 6. Desktop GUI Mockup: Game Browser

2.2.1.5. Scenario Player

This is the sole interface learners interact with while playing KIP's games. It presents the current scenario or stage, accepts user input, and displays the resulting feedback and source references. As gameplay progresses, it updates the interface to reflect new questions, choices, and scenario states according to the rules of the selected game.

2.2.2. Application Layer

The core of KIP is an application layer that houses all of the major systems and coordinates them together. For every presentation layer feature exists a corresponding backend system that must support it.

2.2.2.1. Tech Stack

Java 21 is used to construct KIP's backend, performing all system logic. To store and retrieve platform content and user analytics from the database, Spring Boot is used. It also starts and configures the Java web application.

2.2.2.2. Authentication System

This system manages user login, account creation, guest access, and session validation. It determines whether a user is using KIP as a registered account or as a limited guest, and it controls which features are utilized by KIP based on that status. For guest users, no analytics are recorded, so they can freely try out games in a simplified environment.

2.2.2.3. Game Engine

This system loads and manages the interactive content used during KIP's games. After selecting a game, the learner chooses a topic to focus on, and the system assembles a session from scenarios in the content database that tests that topic. Each scenario defines its own choices,

branches, consequences, and game states, which unfold in sequence based on the selected game and the learner's previous actions.

2.2.2.4. Feedback System

This system provides explanations for the results of a learner's decisions. It identifies whether an answer is correct, incorrect, or context-dependent, explaining the legal reasoning behind the result in order to help a learner understand why a specific choice was safe, risky, or incomplete.

2.2.2.5. Progress Tracker

This system collects the results of each game session and updates a learner's progress accordingly. As the learner engages more with KIP, the progress tracker develops an increasingly accurate estimate of their competency across each topic.

2.2.2.6. Curation System

This system organizes and recommends learning content based on a learner's progress and performance history. It determines what games, modules, and scenarios to prioritize, depending on where the learner needs more practice, while also allowing learners to freely browse and ultimately select any available content.

2.2.2.7. Source Reference System

This system connects game content and feedback to official legal sources, educational resources, and relevant vocabulary. It allows learners to see where KIP's information comes from and, when desired, review the original legal language and documents behind an explanation. It also supports content maintenance by tracking changes to legislation, updating references when needed, and flagging KIP content that may be outdated so administrators can review it, revise it, or disable access to it for the time being.

2.2.3. Data Layer

The features of KIP rely on a large amount of data. Anytime a learner completes a game session, the progress tracker must store the results in the database. This change allows KIP to reevaluate user stats, momentum scores, and recommendations for games and topics. Anytime a learner plays a game, the game engine must be able to request from the database an appropriate sequence of scenarios to be played.

2.2.3.1. Tech Stack

KIP's persistent data is stored in a MySQL database. For local development, Docker is used to run and configure the database environment across our team's systems.

2.2.3.2. Database Structure

KIP uses a single MySQL database composed of related tables for learners, progress, modules, games, scenarios, and legal content.

2.2.3.2.1. User Data

User data stores information associated with individual KIP accounts, including authentication data, profile information, and account status. The application layer uses this data to support authentication and personalization.

2.2.3.2.2. Progress Data

Progress data stores records of a learner's activity and performance across KIP. This includes completed game sessions and scenario attempts, and competency estimates, which are used in tandem to calculate progress statistics that the learner can see.

2.2.3.2.3. Learning Content

Learning content defines the educational structure used during gameplay, including games, scenarios, stages, prompts, answers, and feedback. Each scenario belongs to a game and is associated with the one legal topic it tests. A scenario contains one or more stages, each of which defines a decision point and may link different answers to different subsequent stages to facilitate branching gameplay.

2.2.3.2.4. Legal Content

A piece of legal content contains its user-friendly explanation alongside references to the authoritative sources that support it. Each item is associated with one topic and one or more concepts that comprehension depends on. Each piece of legal content stores the last date that it was updated, which is periodically compared with the last modified date of every legal source referenced by it. When a source has been modified since its associated legal content was last updated, the content is flagged for review so that an administrator can evaluate whether its explanation remains true or requires an update.

2.2.4. Hardware Layer

KIP's hardware layer consists of two end system roles: the server infrastructure on which the application is hosted and the client devices through which users access the application. Because KIP is web-based, users are not required to possess specialized hardware beyond a device capable of running a modern web browser.

2.2.4.1. Server-Side Components

A virtual private server hosts KIP's Java application and supporting services. This server executes backend processing, communicates with the MySQL database, stores application

resources, and responds to requests received from client systems. During local development, Docker is used to supply the database environment, while the publicly-deployed system relies on a server-hosted infrastructure.

2.2.4.2. Client-Side Components

KIP can be accessed from consumer desktop computers, laptops, tablets, and smartphones with a modern web browser and stable internet connection. Client devices are responsible for displaying the web interface, accepting user input, and communicating with the server through network requests.

2.2.4.3. Functions

The hardware layer provides the computing and network resources required to operate KIP. Server-side hardware runs Java application code, processes user requests, accesses and updates persistent data, and delivers website content, while client-side hardware presents the interface and transmits user interactions to the server.

3. Identification of Case Study

This section identifies the groups that KIP is designed to serve and the contexts in which the platform may best be used. It first defines the primary target audience of young adults approaching greater legal independence, then considers secondary stakeholders who may use KIP as a supplemental education resource, and finally outlines additional groups that may benefit from the platform in future applications.

3.1. Primary Target Audience

KIP's primary target audience is young people aged 16–19 who are preparing to enter adulthood and navigate high-risk legal areas, such as driving, work, and substance use. This group is well suited for KIP because its members are old enough for legal consequences to become personally significant, but still young enough to be especially receptive to education, habit formation, and the correction of misconceptions. By focusing on users at this transitional stage, KIP can help build foundational legal awareness and encourage informed decision making before risky habits or misunderstandings become more difficult to change.

3.2. Secondary Stakeholders

The sophisticated automation of learning that KIP is centered around particularly lends itself to self-motivated learners, because it does not require the presence of a third party to guide learning, such as a teacher or parent. However, because the choice of what games and topics to practice is ultimately left to the user, this user could very well be a parent, teacher, or school administrator using KIP as a supplemental educational tool. These stakeholders can benefit from KIP as a way to reinforce legal awareness either inside or outside the classroom.

3.3. Future Users

Future users of KIP may include participants in juvenile diversion programs and community youth groups, as well as students at driving schools. These groups may use KIP as educational resources to improve legal literacy in structured settings.

4. Glossary

4.1. CDC: Centers for Disease Control and Prevention

4.2. KIP: Knowledge Is Power

4.3. MFCD: Major Functional Component Diagram

4.4. UX: User Experience

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